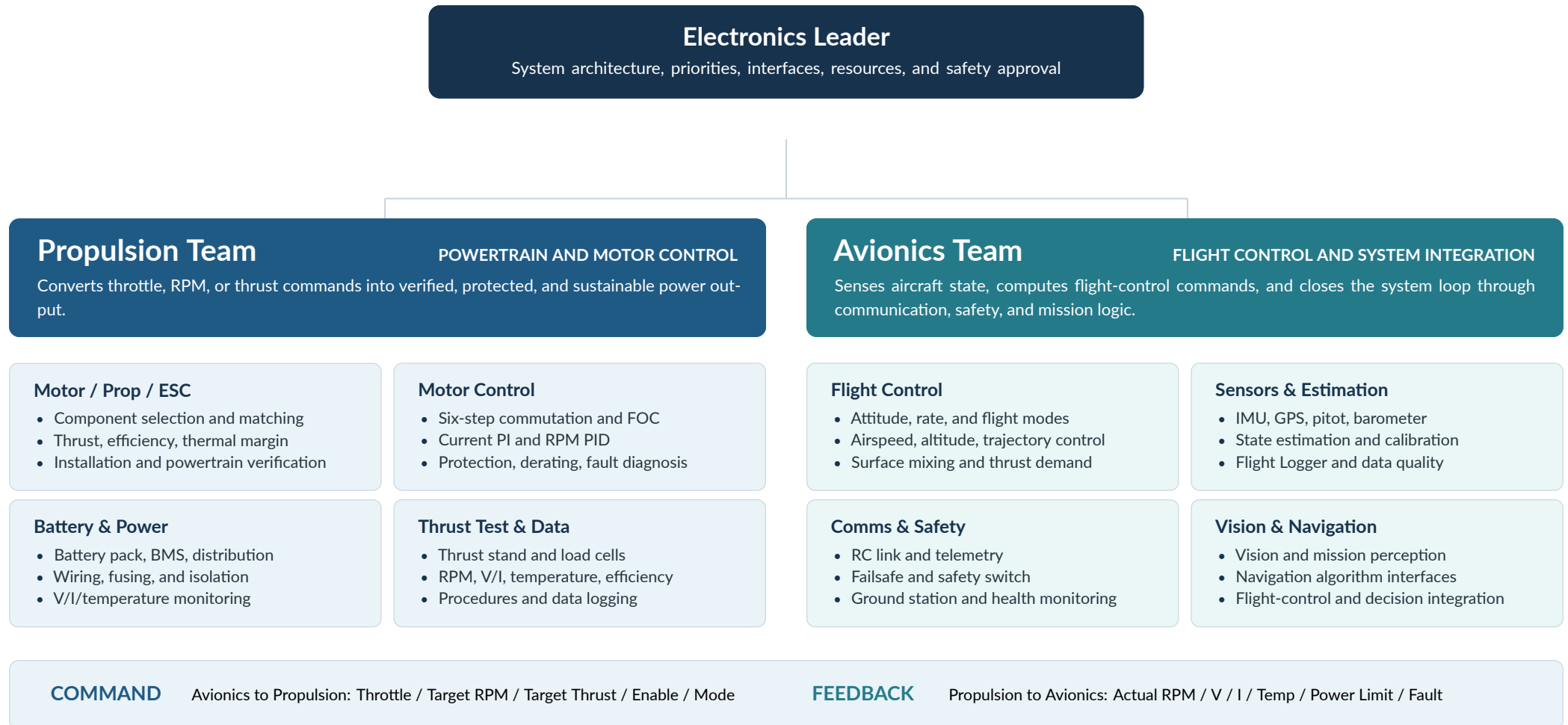


DARES Electronics Team Structure

Fixed-Wing Project: Propulsion and Avionics Responsibilities

Page 1



Responsibility Matrix

Who owns what, what they deliver, and how work is accepted

PROPULSION TEAM

1

Powertrain Design

- Define mission operating points: takeoff, climb, cruise, and maximum continuous power.
- Complete motor, propeller, ESC, and battery matching with thermal-margin analysis.
- Deliver a purchasable BOM, wiring plan, and installation constraints.

2

Motor and ESC Control

- Implement six-step commutation or FOC.
- Own current PI, speed PID, and sensorless or encoder-based estimation.
- Implement overcurrent, undervoltage, overtemperature, stall, and derating strategies.

3

Battery and High-Current System

- Own battery packs, BMS, main power distribution, and connectors.
- Define high-current wiring, fusing, isolation, and maintenance rules.
- Provide reliable voltage, current, and power data to the aircraft system.

4

Test and Validation

- Run thrust, RPM, power, temperature, and efficiency testing.
- Establish repeatable procedures, log formats, and acceptance thresholds.
- Provide validated thrust models to avionics and flight testing.

AVIONICS TEAM

1

Flight Control and Navigation

- Own roll, pitch, and yaw rate control, attitude, airspeed, altitude, and trajectory control.
- Configure ArduPilot, flight modes, control-surface mixing, and mission logic.
- Calculate target throttle, thrust, or RPM from the aircraft state.

2

Sensors and State Estimation

- Integrate IMU, GPS, pitot, barometer, and Flight Logger systems.
- Complete calibration, time synchronization, filtering, and health checks.
- Provide trusted state estimates and fault flags to control functions.

3

Communication and Safety

- Own RC link, telemetry, ground station, and data links.
- Define failsafe, safety-switch, link-loss, and low-battery strategies.
- Define arm/disarm, mode switching, emergency stop, and recovery behavior.

4

System Integration

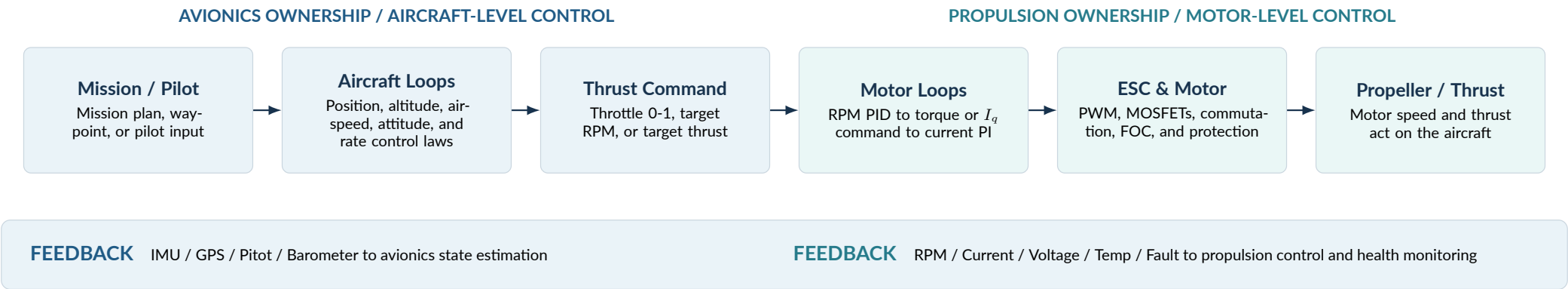
- Manage flight-control and propulsion interfaces, message formats, units, and update behavior.
- Prepare ground integration, HIL/SIL, taxi, and flight testing.
- Integrate vision and perception outputs without duplicating propulsion control.

Core rule: one physical variable has one closed-loop owner. Avionics closes aircraft-state loops; Propulsion closes motor-current, torque, and RPM loops.

PID / PI Control Boundary

Aircraft-level outer loops versus motor-level inner loops

Avionics decides how the aircraft should fly and how much power is required; Propulsion generates that power accurately and safely.



Interface & Collaboration Rules

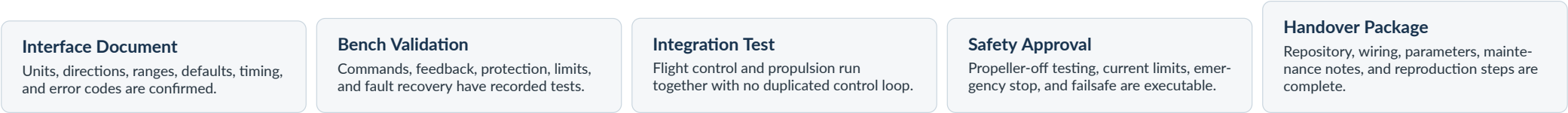
Cross-team interface and acceptance workflow

1. Avionics-Propulsion Interface Contract

Every signal must define its direction, units, valid range, update behavior, and fault handling.

Signal	Direction	Owner	Definition and Fault Behavior
Throttle command	Avionics to Propulsion	Avionics	Normalized 0.0-1.0; clamp out-of-range values; link loss enters the predefined failsafe.
Target RPM / thrust	Avionics to Propulsion	Shared interface	Optional advanced interface; units, limits, and conversion behavior must be defined in the ICD.
Enable / Arm / Mode	Avionics to Propulsion	Avionics + Safety	Define enable conditions, inhibit conditions, emergency stop, reset, and recovery sequence.
Actual RPM	Propulsion to Avionics	Propulsion	Measured value plus a valid flag; used for health checks, not as a replacement for aircraft airspeed.
Bus voltage / current	Propulsion to Avionics	Propulsion	Supports power management, low-energy detection, and overload decisions.
Temperature / limit	Propulsion to Avionics	Propulsion	Report temperatures, derating ratio, current-limit state, and power-limit state.
Fault / heartbeat	Propulsion to Avionics	Propulsion	Report fault code and alive status; Avionics decides the mission-level response.

2. Cross-Team Definition of Done



Weekly cadence: Progress / Plan / Problems. Team leads report system status; project owners report evidence, blockers, and the next test.